

Four-Way ANOVA Report

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Introduction

In this report we are going to write a code that runs a 3-way ANOVA with a 2^k FD for a dataset with 3 parameters A,B,C that affect a certain response in carbonated drinks production.

The Data

As always we insert the data.

```
#|echo: true
Y<-c(45,71,48,65,68,60,80,65,43,100,45,104,75,86,70,96) #each couple of data points refer to
```

We setup the design matrix:

```
D  <- rep(c(-1,1), each = 8)
C  <- rep(rep(c(-1,1), each = 4), 2)
B  <- rep(rep(c(-1,1), each = 2), 4)
A  <- rep(rep(c(-1,1), each = 1), 8)
AB <- A*B
AC <- A*C
AD <- A*D
BC <- C*B
BD <- B*D
CD <- C*D
ABC <- A*C*B
ABD <- A*B*D
ACD <- A*C*D
BCD <- B*C*D
ABCD<- A*B*C*D
#We create the design matrix
```

```
Design <- data.frame(
  Intercept = 1,
  A = A,
  B = B,
  C = C,
  AB = AB,
  AC = AC,
  AD = AD,
  BC = BC,
  BD = BD,
  CD = CD,
  ABC = ABC,
  ABD = ABD,
  ACD = ACD,
  BCD = BCD,
  ABCD= ABCD
)
Design
```

	Intercept	A	B	C	AB	AC	AD	BC	BD	CD	ABC	ABD	ACD	BCD	ABCD
1	1	-1	-1	-1	1	1	1	1	1	1	-1	-1	-1	-1	1
2	1	1	-1	-1	-1	-1	-1	1	1	1	1	1	1	-1	-1
3	1	-1	1	-1	-1	1	1	-1	-1	1	1	1	-1	1	-1
4	1	1	1	-1	1	-1	-1	-1	-1	1	-1	-1	1	1	1
5	1	-1	-1	1	1	-1	1	-1	1	-1	1	-1	1	1	-1
6	1	1	-1	1	-1	1	-1	-1	1	-1	-1	1	-1	1	1
7	1	-1	1	1	-1	-1	1	1	-1	-1	-1	1	1	-1	1
8	1	1	1	1	1	1	-1	1	-1	-1	1	-1	-1	-1	-1
9	1	-1	-1	-1	1	1	-1	1	-1	-1	-1	1	1	1	-1
10	1	1	-1	-1	-1	-1	1	1	-1	-1	1	-1	-1	1	1
11	1	-1	1	-1	-1	1	-1	-1	1	-1	1	-1	1	-1	1
12	1	1	1	-1	1	-1	1	-1	1	-1	-1	1	-1	-1	-1
13	1	-1	-1	1	1	-1	-1	-1	-1	1	1	1	-1	-1	1
14	1	1	-1	1	-1	1	1	-1	-1	1	-1	-1	1	-1	-1
15	1	-1	1	1	-1	-1	-1	1	1	1	-1	-1	-1	1	-1
16	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The Results

We calculate the estimator:

```
X <- cbind(1, A, B, C,D , AB, AC,AD, BC, BD ,CD , ABC,ABD,ACD,BCD,ABCD)
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat
```

```
      [,1]
      70.0625
A      10.8125
B       1.5625
C       4.9375
D       7.3125
AB       0.0625
AC      -9.0625
AD       8.3125
BC       1.1875
BD      -0.1875
CD      -0.5625
ABC      0.9375
ABD      2.0625
ACD     -0.8125
BCD     -1.3125
ABCD     0.6875
```

We see that the analysis matches the results we expected and the results we derived by doing the calculations by hand. We will now do it with a single line of code to calculate p-values as well.

```
df <- data.frame(Y, A, B , C, D)
model <- lm(Y ~ A * B* C * D,data = df)
summary(model)
```

Call:

```
lm(formula = Y ~ A * B * C * D, data = df)
```

Residuals:

ALL 16 residuals are 0: no residual degrees of freedom!

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	70.0625	NaN	NaN	NaN
A	10.8125	NaN	NaN	NaN

B	1.5625	NaN	NaN	NaN
C	4.9375	NaN	NaN	NaN
D	7.3125	NaN	NaN	NaN
A:B	0.0625	NaN	NaN	NaN
A:C	-9.0625	NaN	NaN	NaN
B:C	1.1875	NaN	NaN	NaN
A:D	8.3125	NaN	NaN	NaN
B:D	-0.1875	NaN	NaN	NaN
C:D	-0.5625	NaN	NaN	NaN
A:B:C	0.9375	NaN	NaN	NaN
A:B:D	2.0625	NaN	NaN	NaN
A:C:D	-0.8125	NaN	NaN	NaN
B:C:D	-1.3125	NaN	NaN	NaN
A:B:C:D	0.6875	NaN	NaN	NaN

Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared: 1, Adjusted R-squared: NaN
F-statistic: NaN on 15 and 0 DF, p-value: NA

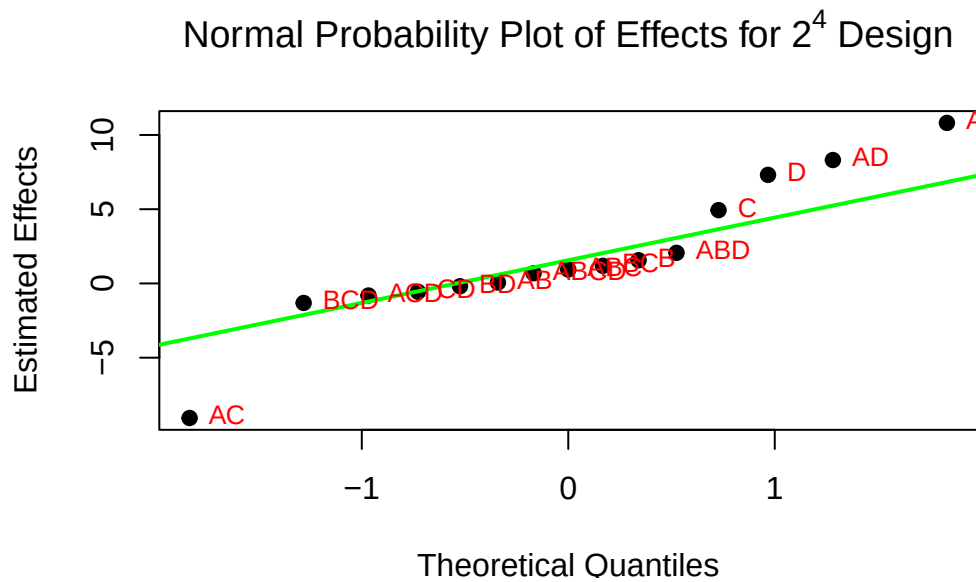
We can see where the problem lies, it cannot calculate errors with only a single response in every level combination. We are going to draw a Normal Probability Plot with the responses estimations instead of the residuals to check which of them are important and which are not:

```
# Factor/effect names (excluding intercept)
effect_names <- c("A", "B", "C", "D",
                  "AB", "AC", "AD", "BC", "BD", "CD",
                  "ABC", "ABD", "ACD", "BCD", "ABCD")

# Sqq plot
qq <- qqnorm(beta_hat[-1],
             main = expression("Normal Probability Plot of Effects for 2"4*" Design"),
             ylab = "Estimated Effects",
             xlab = "Theoretical Quantiles",
             pch = 19, col = "black")

# Add QQ line
qqline(beta_hat[-1], col = "green", lwd = 2)

# Add labels to existing points
text(qq$x, qq$y,
     labels = effect_names,
     pos = 4, cex = 0.8, col = "red")
```



We can confidently deduce that B is not present in any important factor combination, thus we can remove it from our original response vector and repeat the analysis with 1 extra response from B in every level combination!

We repeat

We repeat the analysis but we add a second response for every combination from ignoring the factor B.

```
Design2 <- data.frame(
  Intercept = 1,
  A = A,
  C = C,
  D = D,
  AC = AC,
  AD = AD,
  CD = CD,
  ACD = ACD)
Design2
```

	Intercept	A	C	D	AC	AD	CD	ACD
1	1	-1	-1	-1	1	1	1	-1
2	1	1	-1	-1	-1	-1	1	1
3	1	-1	-1	-1	1	1	1	-1
4	1	1	-1	-1	-1	-1	1	1

5	1	-1	1	-1	-1	1	-1	1
6	1	1	1	-1	1	-1	-1	-1
7	1	-1	1	-1	-1	1	-1	1
8	1	1	1	-1	1	-1	-1	-1
9	1	-1	-1	1	1	-1	-1	1
10	1	1	-1	1	-1	1	-1	-1
11	1	-1	-1	1	1	-1	-1	1
12	1	1	-1	1	-1	1	-1	-1
13	1	-1	1	1	-1	-1	1	-1
14	1	1	1	1	1	1	1	1
15	1	-1	1	1	-1	-1	1	-1
16	1	1	1	1	1	1	1	1

New results

We calculate the new estimator:

```
X <- cbind(1, A, C,D , AC,AD,CD ,ACD)
beta_hat2 <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat2
```

```
      [,1]
      70.0625
A      10.8125
C       4.9375
D       7.3125
AC     -9.0625
AD       8.3125
CD     -0.5625
ACD    -0.8125
```

We see that the analysis matches the results we expected and the results we derived by doing the calculations by hand. We will now do it with a single line of code to calculate p-values as well.

```
df <- data.frame(Y, A , C, D)
model2 <- lm(Y ~ A * C * D,data = df)
summary(model2)
```

Call:

```
lm(formula = Y ~ A * C * D, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.0	-2.5	0.0	2.5	6.0

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	70.0625	1.1842	59.164	7.40e-12	***
A	10.8125	1.1842	9.131	1.67e-05	***
C	4.9375	1.1842	4.169	0.003124	**
D	7.3125	1.1842	6.175	0.000267	***
A:C	-9.0625	1.1842	-7.653	6.00e-05	***
A:D	8.3125	1.1842	7.019	0.000110	***
C:D	-0.5625	1.1842	-0.475	0.647483	
A:C:D	-0.8125	1.1842	-0.686	0.512032	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

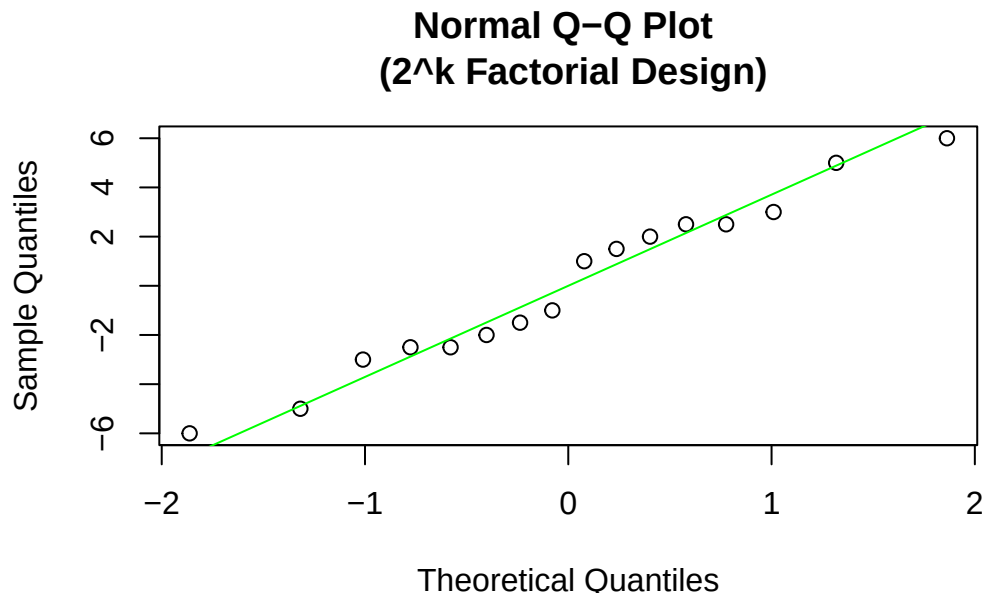
Residual standard error: 4.737 on 8 degrees of freedom

Multiple R-squared: 0.9687, Adjusted R-squared: 0.9413

F-statistic: 35.35 on 7 and 8 DF, p-value: 2.119e-05

To finish the analysis we will do the normal probability plot once again but this time for the residuals:

```
qqnorm(residuals(model2),main = "Normal Q-Q Plot \n (2^k Factorial Design)")  
qqline(residuals(model2),col="green")
```



Once again the residuals seem to follow the normal distribution.

Lenth's Test

Lenth's test will now be used in our original data to overcome the NaN values in the errors, with this test we will overcome this by calculating a pseudo standard error (PSE) to be able to tell which factors are important.

```
# Lenth's Test for Unreplicated  $2^k$  Design

effects <- as.numeric(beta_hat[-1]) # remove intercept
names(effects) <- effect_names      # add names to effects

# Step 1-2: Compute s0
s0 <- 1.5 * median(abs(effects))

# Step 3-4: Keep small effects (< 2.5 * s0) and compute PSE
Es <- effects[abs(effects) < 2.5 * s0]
PSE <- 1.5 * median(abs(Es))

# Step 5: Compute t-values (ratios)
t_values <- abs(effects) / PSE

# Step 6: Critical limits
ME <- 2.658 * PSE # Margin of Error
```



```

SME <- 3.118 * PSE # Simultaneous Margin of Error

# Create results table
lenth_table <- data.frame(
  Effect = names(effects),
  Estimate = round(effects, 3),
  abs_Effect = round(abs(effects), 3),
  t_value = round(t_values, 3),
  Significant = ifelse(abs(effects) > SME, "YES*",
                      ifelse(abs(effects) > ME, "MAYBE", "NO"))
)

knitr::kable(lenth_table, caption = "Lenth's Test for Unreplicated 24 Design")

```

Table 1: Lenth's Test for Unreplicated 2⁴ Design

	Effect	Estimate	abs_Effect	t_value	Significant
A	A	10.812	10.812	8.238	YES*
B	B	1.562	1.562	1.190	NO
C	C	4.938	4.938	3.762	YES*
D	D	7.312	7.312	5.571	YES*
AB	AB	0.062	0.062	0.048	NO
AC	AC	-9.062	9.062	6.905	YES*
AD	AD	8.312	8.312	6.333	YES*
BC	BC	1.188	1.188	0.905	NO
BD	BD	-0.188	0.188	0.143	NO
CD	CD	-0.562	0.562	0.429	NO
ABC	ABC	0.938	0.938	0.714	NO
ABD	ABD	2.062	2.062	1.571	NO
ACD	ACD	-0.812	0.812	0.619	NO
BCD	BCD	-1.312	1.312	1.000	NO
ABCD	ABCD	0.688	0.688	0.524	NO